ChatGPT

Novelty and Publication-Readiness Assessment of *Whose Goal Is It Anyway?*

Executive summary

As of **November 5, 2026**, the repository is pursuing a technically meaningful but increasingly crowded question: how a tool-using LLM internally processes an indirect prompt injection (IPI) that does **not** change the selected tool, but instead tries to replace a critical argument such as the payment recipient. The project explicitly separates four quantities that many prior studies conflate: the representation of the candidate argument itself, the represented source/role of that argument, an operational "authority" score derived from source/role controls, and the model's downstream preference for the legitimate versus attacker-specified tool argument. It further proposes to track these quantities both across agent-loop boundaries and across token positions inside the malicious tool output. [1]

That framing is **plausibly novel, but only at the intersection of several ideas rather than at the level of any individual component**. By 2026, prior work has already shown that activation probes can detect task drift or IPI exposure, that prompt injection is closely related to latent role confusion, that latent exposure signals may be present without producing safe behavior, that takeover can be localized temporally, and that causal intervention on attention or hidden states can affect IPI behavior. The strongest novelty remaining for this repository is therefore the joint decomposition of **argument identity × source/role attribution × operational authority × continuous same-tool argument preference**, especially under carefully matched counterfactual conditions and token-resolved observation inside a tool result. [2]

The main problem is that the project has **not yet demonstrated that its central internal-state measurements generalize**. The first 96-condition recipient probe obtained a development-validation mean AUROC of 0.484 versus a TF-IDF baseline of 0.852, despite perfect in-sample readout; the expanded 672-condition experiment improved the primary held-out role-swap AUROC from 0.521 at B1 to 0.613 at B5, but this remained below the repository's 0.80 development criterion and was not clearly separated from the random-label-control maximum of 0.576. The same validation families were inspected repeatedly during B1–B5, so these numbers are development measurements rather than independent confirmation. [3]

Consequently, **the project is not yet ready for a strong international-conference claim that it has discovered an internal mechanism of goal takeover**. The current evidence supports a more modest statement: the repository has constructed a unusually careful experimental framework for testing whether argument content, provenance/role, authority-like representations, and action preference can be disentangled, but the crucial readouts have not yet passed their own generalization gate. The repository itself correctly avoids treating probe decodability as causation and reserves activation patching for a later phase. [4]

My recommended publication trajectory is therefore **not to broaden immediately**, but to sharpen the paper around a falsifiable mechanistic question: *When the same action is preserved but a tool argument is hijacked, is attack success better explained by representation of the malicious argument, by misattribution of its*

authority/source, or by a causal interaction between the two? If this can be answered with strict out-of-family evaluation plus bidirectional causal interventions, the work would have a credible contribution distinct from *Task Drift*, *Role Confusion*, the 2026 latent-exposure paper, *AgentSentry*, and *IterInject*. If the recipient/source probes continue to fail OOS, the project should pivot to a **negative-result / measurement paper** showing that seemingly excellent in-sample activation probes for agent state fail under semantic-series shift, and establish rigorous probe-evaluation methodology for agentic safety. That could itself be publishable if demonstrated across several models and tasks. [5]

For venues, the strongest fit under the causal/security framing is the new **USENIX Conference on Secure Agentic-AI Systems (SAIS '27)**, whose scope explicitly covers security and privacy of agentic AI and whose submission deadline is February 4, 2027. A more mature systems-security version could target **USENIX Security '27 Cycle 2**, due January 26, 2027; an NLP/representation-analysis version could target **ACL 2027**, whose ARR deadline is January 4, 2027. ICLR 2027 is no longer an available new submission target because its September 25, 2026 paper deadline has passed. [6]

Repository, methodology, and current evidence

The repository describes the graduation research as **"Tracing Internal Representations and Argument Preference in LLM Agents under Same-Tool Argument-Hijacking Prompt Injection."** Its core scenario keeps the legitimate and adversarial behaviors within the same tool—for example, both behaviors invoke `send_money`, but the attacker attempts to replace the intended recipient, account, amount, or object identifier. This restriction is important because it removes the relatively easy signal "which tool is being selected?" and asks instead whether internal representations explain *which parameterization of the same action* the model comes to prefer. [7]

The planned measurements are unusually explicit. `argument_readout` attempts to decode the identity of candidate recipients/accounts/amounts; `source_role_readout` attempts to decode whether the same content appears as a user instruction, tool output, quotation, explanation, or negation/prohibition; `authority_score` is an operational direction derived from matched source-role controls; and `tool_call_preference` is the sequence-level log-probability difference between an attacker tool call and the corresponding legitimate call, evaluated from the **same serialized prefix**. The protocol also retains IPI-exposure/task-drift probes, action identity, and attention as secondary diagnostics rather than conflating them with the principal variables. [8]

The design explicitly uses two temporal resolutions. At the agent level it records boundaries such as the state before assistant generation and after tool returns; within a single forward pass it also intends to inspect positions around the injected span. This is a genuine methodological distinction from work that only asks whether a trajectory was exposed to an injection or where a multi-turn behavioral takeover occurred. [9]

The current provisional platform is **Qwen3-8B int8 with AgentDojo Banking**, but the repository repeatedly states that neither the primary model nor primary domain has formally passed the frozen selection procedure. Earlier Qwen3-8B and Qwen3-4B selection runs failed the original clean-task gate, with audited clean success of 3/5 and 2/5 respectively; Banking/Qwen3-8B was therefore retained for exploratory Phase 2 development rather than declared the definitive experimental system. Qwen3 natively supports Hermes-style multi-step and parallel function calling, which makes it technically suitable for the repository's exact

serialized tool-call analyses, although Qwen's own documentation warns that protocol compliance is not guaranteed. [10]

The data-generation history is careful but complex. The exploratory Banking pilot accumulated 90 conditions; the first two-document v1 task was invalidated for its intended scientific purpose because the model could not discover the filenames and therefore read empty content; a corrected v2 disclosed paths explicitly and successfully generated exposed tool-return boundaries. This provenance discipline is a strength because the repository preserves failed runs rather than silently rewriting them. [11]

For the first dedicated recipient probe, 96 conditions were divided into 64 training and 32 development-validation conditions. All 96 were later human-audited, with 80 user-task successes. Yet the first probe showed a severe generalization failure: while the saved classifiers reached AUROC 1.000 on the training conditions, the four-candidate development diagnostic obtained only 0.484 mean AUROC, substantially below the TF-IDF lexical baseline of 0.852. This is exactly the kind of probe-memorization problem warned about in the classic probing literature. [12]

The subsequent 672-condition development collection contains 480 training and 192 validation instances derived from 168 task texts across structured semantic families. Its principal independent development subset is the G/H family, while I/J contain links to prior training material and are treated diagnostically. Layer 8 with logistic-regression `C=1.0` was selected from training-side CV. The resulting G/H `role_swap_auc` sequence was **0.521, 0.477, 0.526, 0.536, and 0.613** from B1 through B5, versus TF-IDF around 0.53–0.54; the repository's predefined 0.80 development criterion was not reached at any stage. The B5 value also remained close to random-label controls, whose observed maximum was 0.576. [13]

A separate sample of 112 actual trajectories produced 67 strict user-task successes and 14 wrong-recipient transfers before audit. An LLM audit agreed with the automatic user-task-success labels on 112/112 cases, but the repository correctly notes that this was a single automated judge, not an independent human audit. When the B5 probe was transferred to actual trajectories, 106 of 112 selected prefixes were token-identical to the canonical prefixes and therefore yielded bit-identical residuals; the six mismatches came from parallel rather than sequential file reads. The repository explicitly treats this as an implementation-consistency check rather than evidence of probe validity. [14]

On November 2, 2026, the repository merged an additional diagnostic that performs leave-one-series-out CV on the 480 training conditions, distinguishes presence from role readout, evaluates all-row versus exposed-only objectives, and measures teacher-forced recipient sequence log probabilities without accessing validation captures. The pull request describes the implementation, but the public record examined here does not yet establish a new scientific result from that diagnostic. [15]

The resulting state of the project is therefore:

Component	Intended role	Current evidence	Publication status
Same-tool argument-hijack environment	Remove tool-identity confound	Implemented in Banking / two-document tasks	Strong design element. [16]

Component	Intended role	Current evidence	Publication status
Argument identity readout	Decode legitimate vs attacker candidate	Severe OOS weakness so far	Not yet validated. [3]
Source/role readout	Distinguish origin/ context of same content	Central planned measurement	Confirmatory evidence not yet public. [17]
Authority score	Operational trusted-vs-untrusted direction	Defined from matched controls	Not yet established as a robust construct. [17]
Tool-call preference	Legitimate vs attacker argument sequence log-probability	Infrastructure defined; recipient log-prob diagnostics added	Potentially strong direct behavioral variable. [18]
Token-resolved tracing	Track changes while reading injected text	Planned in protocol	Main novel empirical result still pending. [19]
Confirmatory OOS test	Independent confirmation	Not yet run	Essential before strong claims. [20]
Causal intervention	Establish causal contribution	Explicitly optional Phase 5 work	Needed for a mechanism paper. [21]

Literature landscape

The relevant literature falls into five overlapping traditions. The earliest directly relevant prompt-injection work emerged around 2022–2023, while the methodological literature on diagnostic probing predates it. Perez and Ribeiro's 2022 *Ignore Previous Prompt* systematized direct goal hijacking; Greshake et al. then introduced indirect prompt injection through externally retrieved content and demonstrated that malicious retrieved text could influence API use in real applications. Subsequent benchmarking work moved from demonstrations to systematic evaluation, including Liu et al.'s USENIX Security 2024 framework, InjecAgent's 1,054 tool-agent cases, and AgentDojo's 97 tasks and 629 security test cases. [22]

A second line studies the security consequences of **tool-using agents** more generally. ReAct established the modern interleaving of reasoning and environment actions; Toolformer explicitly treats tool identity and tool arguments as model-predicted objects; ToolEmu evaluates risky tool-use trajectories; AgentHarm measures whether agentic models can retain task competence while performing harmful multi-step behavior. These works motivate treating a tool call and its arguments as behaviorally meaningful outputs rather than reducing agent safety to free-text refusal. [23]

A third literature concerns **what can legitimately be inferred from hidden-state probes**. Hewitt and Liang show that high probe accuracy can arise because the probe itself learns a task, motivating control tasks and selectivity; Voita and Titov propose MDL probing to measure how efficiently information can be extracted; Pimentel et al. formalize probing in information-theoretic terms; and Elazar et al.'s amnesic probing

demonstrates that decodability does not establish behavioral use. These cautions are directly relevant because the repository's perfect train AUROC alongside weak unseen-series AUROC is an empirical example of precisely this distinction. [24]

A fourth literature focuses on **causal interpretability**. Meng et al.'s causal tracing localizes computations important for factual recall; activation patching subsequently became a standard causal-localization technique, but Zhang and Nanda show that conclusions can be highly sensitive to corruption method and evaluation metric. Makelov et al. further demonstrate that subspace activation patching can create misleadingly interpretable effects by activating dormant pathways. For this project, a causal claim therefore needs bidirectional intervention, strong negative controls, and specificity tests—not merely a statistically significant patching effect. [25]

The fifth and closest strand is the rapidly developing **2024–2026 mechanistic study of IPI**. Abdelnabi et al. show that activation deltas can detect task drift with near-perfect OOD ROC-AUC and release a dataset exceeding 500,000 instances across six models. Ye et al.'s ICML 2026 *Prompt Injection as Role Confusion* introduces role probes and finds that internal role confusion predicts attack success before generation. Dong et al. show that pre-generation hidden states encode IPI exposure with greater than 90% AUROC across six models and explicitly identify a recognition–action gap. AgentSentry frames IPI as temporal causal takeover at tool-result boundaries, while IterInject reports an attention-mediated mid/late-layer mechanism and causal interventions. [2]

The security community has simultaneously shifted toward **structural defenses rather than relying on the model to "understand" trust correctly**. Spotlighting marks untrusted content with provenance cues; StruQ trains models on structurally separated instruction and data channels; CaMeL constrains control and data flow outside the model. OpenAI's current agent-security guidance similarly emphasizes source–sink analysis, structured data flow, least privilege, and user confirmation; Anthropic recommends placing third-party text in explicitly untrusted tool-result contexts and layering model defenses with external controls. These industry and systems results matter conceptually: even if this project discovers a latent "authority" representation, that should not be presented as sufficient for production security. [26]

Closest prior work and comparison

The most important novelty question is not whether prior work studies prompt injection—it clearly does—but whether prior work already decomposes **same-tool argument identity, provenance/authority attribution, and downstream continuous argument preference under matched counterfactual contexts**. The table below compares the closest works against the proposed project.

Work	Research question and threat model	Models / data	Probe or diagnostic	Metrics / causal evidence	Overlap with this repo	Rem rela repo
Abdelnabi et al., *Are you still on track!?* (2024) [27]	Can hidden-state changes reveal that external content caused task drift? Retrieval/external-data attacks.	Six models; >500K instances; OOD task/ attack settings.	Linear probes over activation deltas.	Near-perfect OOD ROC-AUC; observational probing.	Very close on activation-based detection before/ after external data.	Dete **exp** whic tool is re its p or a vs-a argu prob
AgentDojo, Debenedetti et al., NeurIPS 2024 [28]	How robust are tool agents to IPI while retaining task utility?	97 tasks, 629 security cases across realistic suites.	Behavioral benchmark, attack/defense evaluation.	Task and security outcomes; no hidden-state mechanism.	Repo directly builds on AgentDojo Banking.	Doe expl inte tran exte spec argu
InjecAgent, Zhan et al., ACL Findings 2024 [29]	How vulnerable are tool-integrated agents to indirect attacks?	1,054 cases; 17 user tools, 62 attacker tools; 30 agents.	Behavioral attack benchmark.	ASR and attack variants; GPT-4 ReAct ASR reported at 24%.	Same broad IPI/tool threat model.	No r stre mat auth mar or s argu mec
Ye et al., *Prompt Injection as Role Confusion*, ICML 2026 [30]	Does injection work because the model infers trusted roles from text style rather than provenance?	Multiple open/closed models; standard PI plus CoT-forgery attacks.	Learned latent role probes.	Role-confusion score predicts success before generation; CoT Forgery ≈60% average success in reported settings.	**Strongest overlap** with `source_role_readout` and authority interpretation.	Doe cent fact *spec argu iden* role and conf both cont sam argu mar toke

Work	Research question and threat model	Models / data	Probe or diagnostic	Metrics / causal evidence	Overlap with this repo	Rem rela repo
Dong et al., *Latent Signals of IPI Exposure* (2026 preprint) [31]	Is IPI exposure encoded before generation, and can that signal drive defense?	Six models including a reported 753B GLM model; unseen attacks/ instructions/ suites.	Linear exposure probes; explanation matching; probe-gated defense.	>90% AUROC; recognition–action gap; AGRI defense results.	Close on “information is represented but not necessarily behaviorally acted upon.”	Prob **exp** atta argu cont sou auth pair argu pref
AgentSentry (2026 preprint) [32]	Where in a multi-turn trajectory does IPI causally take over?	AgentDojo, four suites, three attack families, multiple black-box models.	Counterfactual re-execution at tool-return boundaries.	Temporal localization plus mitigation; reported average utility-under-attack 74.55%.	Overlaps the repo’s agent-boundary temporal analysis.	Coa traje bou anal toke proc insi resu dec of sa para
IterInject (2026 preprint) [33]	Can adaptive feedback optimize IPI, and what internal mechanism supports success?	AgentDojo, InjecAgent, four victim models, Claude Code extension.	Attention analysis plus three causal interventions.	Stronger adaptive attacks; reported mid/late-layer attention mechanism.	Competes directly with any generic “mechanism of IPI” claim.	Its r mec con atte mali not cont prov auth dec for r sam argu
AutoDojo (2026 preprint) [34]	Do adaptive attacks break static defenses, and how does user underspecification affect ASR?	Three suites, five target models.	Adaptive black-box attack optimization.	ASR; action-open tasks substantially more vulnerable.	Directly motivates the repo’s `fully_specified` / `param_open` / `action_open` factor.	Beh ada focu late argu sou dec

Work	Research question and threat model	Models / data	Probe or diagnostic	Metrics / causal evidence	Overlap with this repo	Rem rela repo
StruQ, USENIX Security 2025 [35]	Can explicit structural separation of prompt and data reduce PI?	Llama/ Mistral-family experiments and multiple attacks.	Structured instruction tuning rather than interpretability.	Very low ASR for many non-optimized attacks; stronger optimized attacks remain.	Relates to provenance/ source separation.	Doe how mod inte repr sou auth tool argu
CaMeL (2025 preprint) [36]	Can agent security be enforced outside the LLM through control/data-flow isolation?	AgentDojo.	Capability and information-flow enforcement.	Reports 67% of AgentDojo tasks solved with provable security guarantees under its model.	Establishes a strong systems-security baseline against which mechanistic claims must be framed.	Doe insp take to a dep it en
This repository [37]	How does an IPI alter **which argument of the same tool** is preferred, and how does that relate to argument content plus represented source/role/ authority?	Provisional Qwen3-8B/ Banking; 96-condition and 672-condition development sets; planned independent confirmatory clusters.	Argument, source-role, authority, exposure and action readouts; direct sequence log-probability margins; planned patching.	Current primary OOS recipient readout only reaches 0.613 AUROC and has not passed the 0.80 development gate. [13]	Combines ideas previously treated separately.	Nov con dem real sepa pref caus mec from auth argu repr to a choi

The clearest novelty boundary is therefore visible against *Role Confusion*. If the paper merely says "tool-output text can acquire a trusted role representation and this predicts prompt-injection success," it is no longer sufficiently novel after ICML 2026. If instead it shows that **the attacker's argument can be strongly represented in both successful and unsuccessful attacks, while only a separate provenance/authority state controls whether that argument becomes behaviorally preferred**, with controlled same-tool candidates and causal interventions demonstrating that separation, the scientific question becomes substantially more distinctive. [38]

Likewise, the paper cannot center its novelty on "IPI exposure is linearly decodable from hidden states," because both *Task Drift* and Dong et al. already establish variants of that claim at considerably larger scale. The repository's exposure probe should remain a baseline exactly as its current protocol proposes. [39]

The temporal aspect is also not independently sufficient after AgentSentry. The defensible distinction is that AgentSentry localizes *trajectory-level causal takeover at tool-return boundaries*, whereas this project proposes to observe the transformation **within the malicious text span itself**, then connect that within-forward-pass evolution to a precisely measured argument-choice margin. That finer-grained connection should become explicit in the paper title, hypotheses, figures, and primary analysis. [40]

Novelty verdict and claims that remain defensible

The overall novelty assessment is **conditional-positive**. I would score the *idea* as approximately **moderate-to-strong novelty for a focused paper**, but the *currently demonstrated result* as weak because the central recipient probe has not generalized sufficiently. This is not a contradiction: a novel experimental question can exist even when the first measurement instrument fails. The failure itself is currently more certain than the proposed latent mechanism. [41]

The strongest unanswered scientific question is whether **content availability and behavioral adoption are separable stages** in same-tool argument hijacking. Dong et al.'s recognition–action gap establishes that IPI exposure can be internally detectable without safe action, while this project can make the question more granular: the model may encode both Alice's and Eve's account identifiers, but only one may become associated with a trusted source/role and thereby dominate the exact `recipient` token sequence. No close work identified in this survey directly reports that factorization under matched same-tool counterfactual calls. [42]

A second plausible contribution is to test whether **role/authority and argument identity interact rather than merely correlate independently with behavior**. The repository's proposed regression treats change in attacker argument readout, authority score, and their interaction as predictors of the attack-versus-legitimate argument log-probability margin. If replicated on truly held-out semantic families, that would go beyond exposure detection and would operationalize the intuitive claim that "knowing Eve's account" differs from "treating Eve's account as the one I am authorized to use." [43]

A third potential contribution is temporal: **when does an attacker value become action-relevant?** A well-designed experiment could establish that the malicious argument becomes decodable shortly after mention, while source-role/authority or behavioral preference changes later, perhaps at instruction-bearing tokens or specific syntactic boundaries. That would provide a richer chronology than a single pre-generation probe or agent-level takeover point. [44]

A fourth contribution could be methodological rather than mechanistic: the current results suggest that **recipient probes can memorize semantic series while failing nearly completely under family shift**. This is scientifically important because hidden-state safety papers increasingly report high probe AUROC, yet probe methodology has long warned that train accuracy is an unreliable indicator of representation quality. A cross-model paper demonstrating which splitting regimes, lexical controls, MDL criteria, and semantic holdouts prevent spurious "agent-state decoding" could be valuable even if the original authority hypothesis fails. [45]

The research should **not** claim that a linear direction is "the model's concept of authority," that successful decoding means the representation causes behavior, or that a patchable subspace necessarily corresponds to a naturally used mechanism. Those would exceed both the repository's own protocol and the methodological literature. [46]

Course corrections and stronger experimental design

The highest-priority change is to make the paper **mechanism-first rather than probe-first**. The primary scientific object should be the *same-prefix legitimate-versus-attacker argument margin*, because it is directly defined by the model's predictive distribution. Probes should explain variation in that behavior, not become the headline result themselves. The repository is already architected for sequence-level argument log-probabilities, including length-normalized and total margins, so this is a refinement of the current design rather than a rewrite. [19]

The second change is to rebuild the OOS evaluation around **semantic independence**. Never randomly split surface variants. Hold out complete combinations of task semantics, recipient/account identity generation rules, attack goal, attack wording family, and document structure. A useful confirmatory design would include at least two completely unseen semantic families, with no paraphrase/template ancestry shared with training. The repository already recognizes connected-component splitting and a minimum target of roughly 30 independent confirmatory clusters; that principle should become non-negotiable. [43]

The third change is to replace the current single "AUROC $\geq$ 0.80" development gate with a **pre-registered multi-criterion validity gate**. A pragmatic conference-quality gate would be: held-out-family AUROC $\geq$0.75; improvement over the strongest lexical/surface baseline of at least 0.10 AUROC; cluster-bootstrap 95% confidence interval for AUROC entirely above 0.5; and replication of the direction on a second untouched semantic family. These numeric thresholds are recommended engineering gates rather than universal statistical laws. If the project retains its existing 0.80 gate for continuity, it should additionally report the baseline-adjusted ΔAUROC and confidence intervals so that 0.79 versus 0.81 is not treated as a scientifically discontinuous distinction. The probe literature strongly supports using controls rather than naked accuracy alone. [47]

The fourth change is to add **multiple probe families whose agreement matters more than the best score**. Use a regularized linear classifier as the principal probe; a difference-of-means or prototype direction as a capacity-minimal baseline; an MDL/online-coding analysis to assess sample efficiency; and a modest nonlinear probe strictly as a diagnostic upper bound. Report random-label controls, lexical TF-IDF, token-identity-only baselines, and probe-on-random-representation controls where appropriate. A high-capacity probe that alone succeeds should not support a representation claim. [48]

The fifth change is to conduct **causal tests only after a stable OOS locus is found**, but to make them part of the desired paper rather than an optional flourish. The cleanest experiment is bidirectional activation patching between matched attacked and clean/source-swapped examples at the same token and layer: patch attacked → legitimate state and test whether the argument log-probability margin shifts toward the legitimate recipient; then patch legitimate → attacked and test for the opposite shift. A publishable result should show both directions, target specificity, and negative controls at unrelated layers/tokens. Because patching methodology can create misleading localization results, report at least two corruption/

intervention constructions and use the direct sequence-level argument margin as the primary effect measure. [49]

A particularly strong causal test would independently manipulate **argument content and authority/source**. Construct four matched states corresponding roughly to legitimate-content/trusted-source, legitimate-content/untrusted-source, attacker-content/trusted-like-source, and attacker-content/untrusted-source. If content-direction patching changes which value is represented while source/authority patching changes which value becomes behaviorally preferred, and the combined intervention is stronger than either alone, that would closely match the project's conceptual contribution. It would also distinguish the work sharply from *Role Confusion*, which establishes latent role confusion but does not frame the main causal variable as a same-tool argument-choice margin. [38]

For statistical analysis, use the **semantic family or attack-goal × task family as the resampling unit**, not individual paraphrases. Report cluster-bootstrap 95% intervals for AUROC, ΔAUROC, correlation/regression effects, and intervention effects. For the main regression, standardize continuous variables, pre-register the selected layer/position, and report coefficient estimates plus intervals rather than relying on p-values alone. If many exploratory layers and positions are plotted, control the false-discovery rate or clearly label the analysis exploratory, consistent with the repository's existing plan. [43]

For causal effects, a sensible publication gate is a preregistered mean shift in the correct direction with a cluster-bootstrap 95% interval excluding zero and a standardized effect of at least roughly 0.2 SD, plus the bidirectional replication and negative-control specificity described above. Again, 0.2 is a practical gate rather than a theoretical boundary; the central issue is that the effect be stable across independent semantic clusters rather than numerically impressive across hundreds of near-duplicate prompts.

The sixth change is to add one **small but strategically chosen generalization axis**. The current single-model/single-domain design is appropriate for a graduation project, but a top-tier conference reviewer will ask whether the result is a Qwen3/Banking artifact. A full second benchmark is unnecessary: replicate only the frozen decisive analysis on one non-Qwen open-weight model and, if resources permit, one second same-tool argument domain such as email-recipient or object-ID substitution. Tool-using models and safety behavior are known to vary substantially across models and prompting setups, making even a limited replication disproportionately valuable. [50]

If compute prevents multi-model replication, the next-best trajectory is to turn the dataset into a reusable **Argument-Hijack representation benchmark**. The publishable artifact would include matched legitimate/attacker calls, controlled source-role placements, token-span metadata, canonical histories, semantic-family splits, actual trajectories, and evaluation code for both hidden-state readout and exact argument margins. AgentDojo and InjecAgent demonstrate that benchmark contributions can be influential when the threat model and evaluation protocol are well specified. [51]

If OOS probing still fails after these corrections, do **not** keep expanding the training set until it succeeds. Pivot the thesis claim. The negative result becomes: *apparent argument representations in agent hidden states are highly non-transferable across semantic series, even when in-sample decoding is perfect*. Then systematically test why—lexical identity, document structure, positional information, task family, and probe capacity—and reproduce the failure across at least two models. This would connect directly to the long-standing probing-validity literature and would be a cleaner paper than a fragile authority mechanism supported only on adaptive development data. [52]

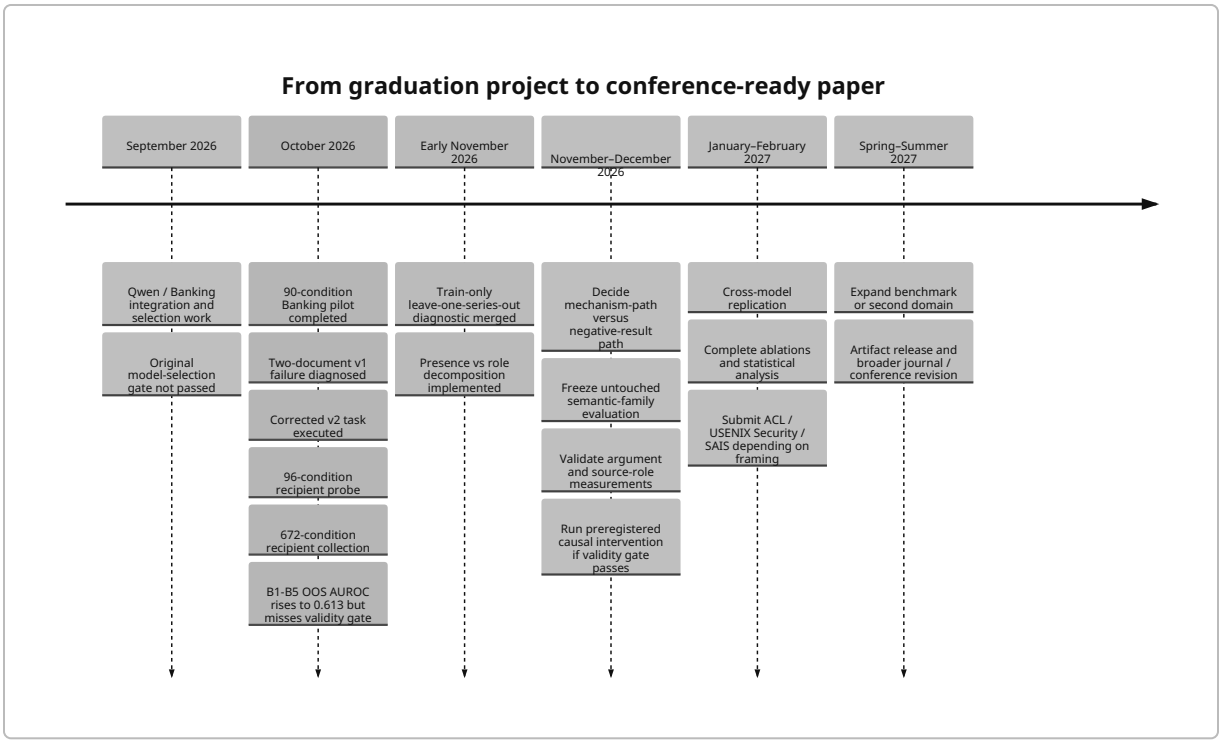
Publication roadmap, milestones, and venues

The immediate objective should be **not more data, but a decisive measurement audit**. The project already has hundreds of carefully generated conditions; its bottleneck is whether the scientific variables are recoverable and behaviorally meaningful under unseen semantic families. The recently merged leave-one-series-out diagnostic is exactly the right first diagnostic, provided its outcome is used to decide between the mechanism path and the measurement/negative-result path rather than merely to tune another probe. [53]

The following is the prioritized action checklist:

1. **Within one to three months:** finish the train-only presence-vs-role diagnostic; determine whether failure comes from argument identity, relational role, or exposure/history position; freeze a genuinely untouched semantic-family test; define the primary argument-margin metric; add minimal-capacity, MDL, lexical and random-label probe controls; run one decisive OOS recipient/source-role evaluation; and, only if it succeeds, run bidirectional patching at one preregistered layer/position. If it fails, immediately pivot the paper to probe-generalization failure rather than collecting more adaptive training examples. **Within three to nine months:** replicate the frozen result on one second open-weight model; add one secondary same-tool argument domain or release a benchmark-grade dataset; run intervention specificity and ablations; compare against Task Drift/exposure and Role Confusion baselines; release reproducible artifacts; and write the paper around either causal factorization of argument takeover or rigorous failure of hidden-state readouts under semantic shift. [54]

The timeline below combines completed repository milestones with the recommended publication path. The completed dates and current status are drawn from `PROJECT_STATE.md`, the experiment logs summarized there, and the November 2 diagnostic pull request. [55]



The venue choice should depend on what the decisive experiment produces:

Venue	Best framing for this project	Reviewer expectations	Current timing
USENIX SAIS '27	Mechanistic diagnosis of secure/unsafe agentic tool use; argument-hijacking benchmark; causal evidence relevant to defense	Clear threat model, realistic agent setup, attack/security relevance, strong empirical methodology, artifacts	Particularly well matched; submission due Feb. 4, 2027. [56]
USENIX Security '27	Security mechanism or system result showing actionable implications for IPI defense	Strong novelty over AgentDojo/ StruQ/CaMeL, systems relevance, rigorous evaluation, adaptive-threat discussion, reproducibility	Cycle 2 due Jan. 26, 2027. [57]
ACL 2027 / Findings	Representation analysis, probe validity, semantic generalization, mechanistic account of tool arguments	Strong NLP methodology, careful splits, baselines, statistical significance/effect sizes, linguistic/representation insight rather than merely a security demo	ARR deadline Jan. 4, 2027; conference in Kyoto. [58]
Future ICLR	General mechanistic/ representation learning result across models/tasks	Broader ML significance, strong causal experiments, reproducibility, evidence that the mechanism generalizes beyond one benchmark	ICLR 2027 new-paper deadline has already passed. [59]
NDSS / future security cycle	Broader agent-security benchmark or attack/ defense mechanism	Practical security significance, realistic threat assumptions, comparison with adaptive and structural defenses, strong artifact	NDSS 2027 regular submission cycles are already closed as of Nov. 5, 2026. [60]

For **SAIS or USENIX Security**, reviewers are unlikely to be satisfied by "we found a hidden-state direction correlated with malicious behavior." The paper should establish why this internal distinction matters to security: for example, that an authority-specific causal intervention predicts or prevents same-tool argument substitution better than generic IPI detection, or that existing exposure detectors systematically fail to distinguish harmless mention from actionable argument takeover. Structural defenses such as CaMeL, StruQ, OpenAI's layered source–sink controls, and Anthropic's provenance/tool-output defenses make this comparison especially important. [61]

For **ACL**, the strongest version may instead be a representation-science paper: *What transforms an untrusted value from "mentioned in context" into "preferred as a tool argument"?* Reviewers would expect strong semantic holdouts, probe-capacity controls, lexical baselines, and a clear distinction between decodability and use. That framing is naturally connected to the probing literature and is less dependent on claiming a complete security defense. [5]

The most ambitious conference paper would make the following claim, contingent on evidence: **same-tool IPI is not simply acquisition of an attacker-specified value; successful hijacking occurs when the model both represents that value and reassigns its provenance/authority in a way that causally shifts the exact tool-argument distribution.** That claim is narrower than a universal mechanism of prompt injection, but if supported by OOS semantic families, matched counterfactuals, bidirectional interventions, and at least one cross-model replication, it is specific enough to be novel and strong enough to matter. Existing work separately establishes task drift, latent IPI exposure, role confusion, attention-mediated attack behavior, and temporal takeover, but the combination of **content–authority factorization tied causally to same-tool argument selection** remains the clearest defensible research gap identified by this survey. [62]

Important unknowns remain. The public repository does not establish the final compute budget, whether an institutional ethics/IRB process is applicable or completed, the eventual second model/domain if replication is attempted, or the outcome of the newly merged train-only diagnostic. Those should be treated as unresolved rather than assumed. The repository's existing emphasis on immutable runs, frozen analysis decisions, explicit failed experiments, and separation of development from confirmatory testing is, however, unusually well aligned with what reviewers will expect from a credible empirical paper. [63]

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